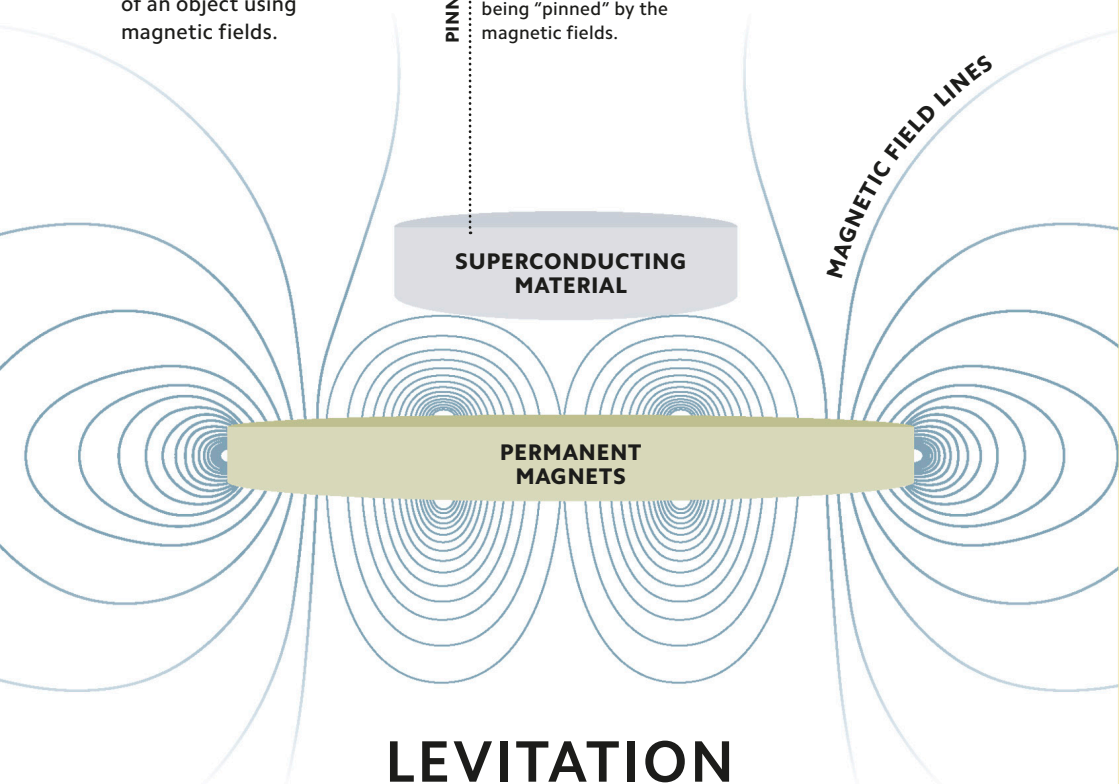


Quantum lift

Superconductors may be used to achieve magnetic levitation: the suspension of an object using magnetic fields.

PINNED IN PLACE The superconductor repels the magnetic fields of the permanent magnets. This can result in stable levitation due to the superconductor being “pinned” by the magnetic fields.



LEVITATION

Superconductivity is a phenomenon observed in special materials, usually at very cold temperatures near absolute zero ($-459.67^{\circ}\text{F}/-273.15^{\circ}\text{C}$), in which electrical resistance vanishes (see p.81). Superconducting electromagnets are made from superconducting wire coils that conduct massive currents, generating powerful magnetic fields. When a superconductor is in a magnetic field, small currents are created. These currents then create opposite magnetic fields, expelling the field and preventing the magnetic field from penetrating the superconductor.